

EEET202

INES & TRANSFORMERS

MODULE:1

- Equalizer Rings & Dummy Coils

edge about constructional details of DC machines

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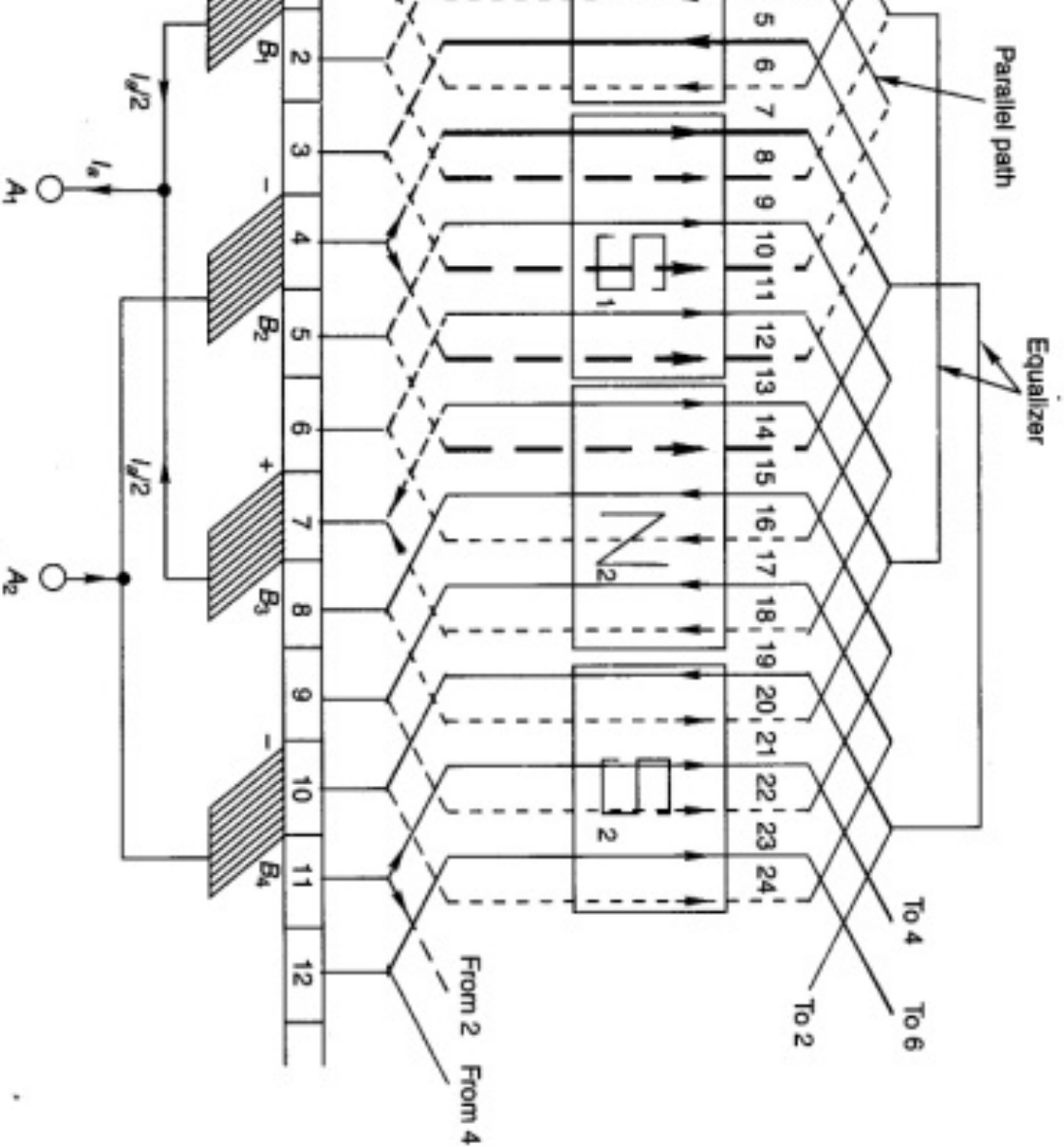
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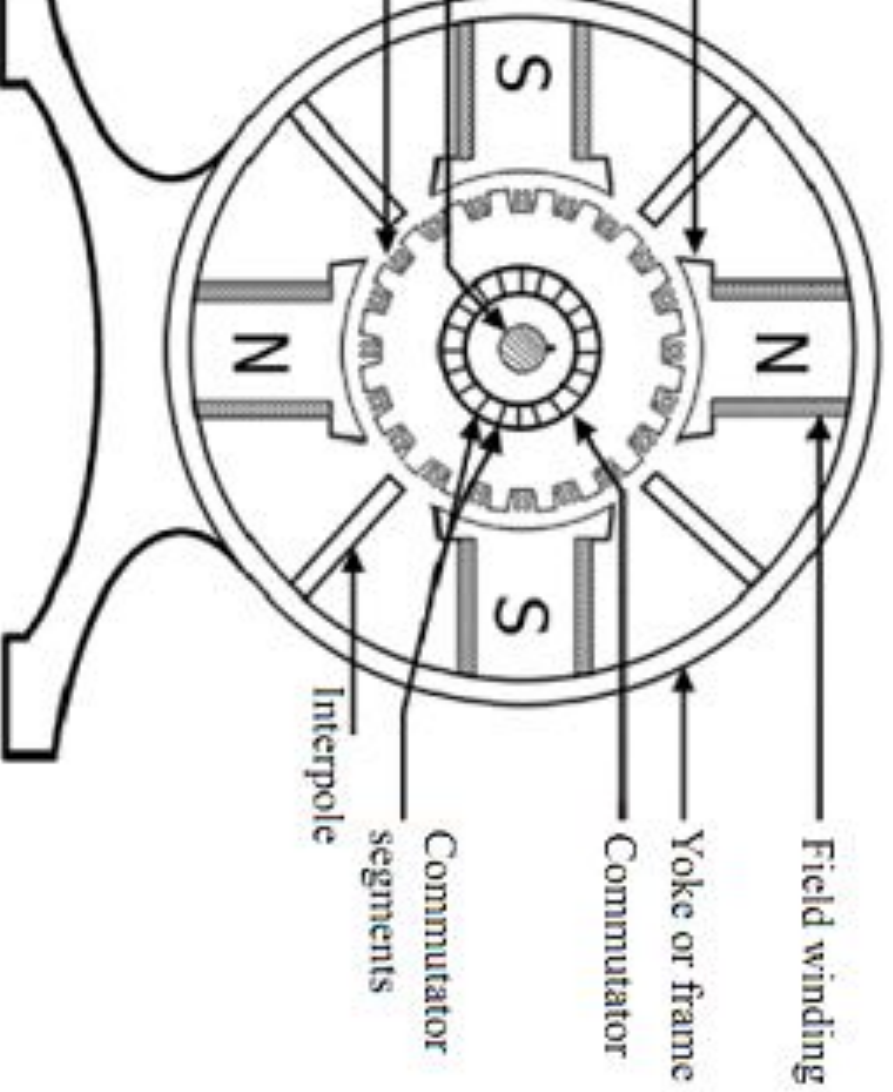


Equalizer Rings



for 4-poles, 12-slot armature (single-turn coils, 2 coil-sides/slot)

DC Machine



Equalizer Rings

The number of parallel paths is equal to the total number of poles. In a machine with equalizer rings, the flux from all the poles is exactly the same, hence the current carried by each path will be equal.

Under normal operating condition, there are variations in air-gap length due to the magnetic properties of pole material which causes an unequal distribution of induced emf in the various parallel paths.

Under a load condition, the armature conductor under stronger poles generates greater emf and carries more current than the conductor under weaker poles.

Some brushes are overloaded and carry more current than others, while the remaining brushes carry less current due to the unequal distribution of current.

Equalizer Rings

can't reverse completely in a given commutation time
overloaded brush and that will cause sparking between
commutator surface.

Equalizer rings are used to overcome such problem.

With multiple winding, there is only 2 parallel paths for any number
of poles. Equalizer rings ensure that the current in all the parallel
paths are distributed under all the poles.

Equalizer rings do not solve the issue of unequal current distribution and no
arcing.

Dummy Coils

It is possible only with particular number of conductor or dummy coils. If the number of slots available in the armature winding does not meet the requirement of the winding. (It means that the available slots are more than the required number of conductors).

Dummy coils are employed without armature winding in that case and dummy coils are employed in that slots.

Dummy coils are similar to other coils except their ends are cut, short circuited and do not connect with the commutator bars.

Dummy coils are simply **to provide mechanical balance for the**

armature. When connect with commutator bars they do not affect the characteristics of the winding.

Dummy Coils

At the wave winding of 26 slots and 4 – poles.

winding

$(P / 2)$

$/ 2$

an even integer therefore the winding is not
inductors. However if the number of coils are 25

$(P / 2)$

$/ 2$

coil ($26 - 25 = 1$) is inactive, it is called as dummy